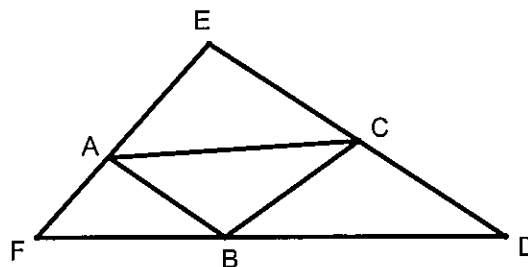


8. Let f be a real-numbered function. If $f(x) = \frac{\log(x-2)}{x^2-40} + \sqrt{8\sqrt{3}-x}$, then the domain of f ,

given in simplified radical form in interval notation, is $(k, a\sqrt{b}) \cup (c\sqrt{d}, g\sqrt{h}]$ where k , a , b , c , d , g , and h are positive integers. Find the value of $3k+2a+b+c+d+7g+h$.

9. In the diagram, points A , B , and C lie on $\overline{FE}$, $\overline{FD}$, and $\overline{ED}$ respectively. If $EA=10$, $AF=5$, $BF=12$, $BD=22$, $CD=16$, and $CE=19$, find the area of $\triangle ABC$. Express your answer as an improper fraction reduced to lowest terms.



10. Find the degree of the following polynomial: $3x(x^3+5)+6(x^2(x-7)^3)^2+34x-9$.

11. There are 12 teams in a league. Each team in the league must play every other team in the league exactly 10 times in one season. Find the number of games that will be played among the teams in this league during one season.

12. Find the sum of all distinct positive values of y that satisfy the system:
$$\begin{cases} \log_9 x = \log_3 y \\ x^2 - 5y^2 = -6 \end{cases}$$

13. The first term of an arithmetic sequence is 8, and the thirteenth term of this arithmetic sequence is -28 . Find the **sum** of the other eleven terms of this arithmetic sequence.

14. In the diagram, $AB=50$ and $BC=40$. If
$$\frac{\sin(0.5(\angle CAB - \angle CBA))}{\cos(0.5(\angle ACB))} = \frac{12}{25}$$
, find the exact length AC .

